# The Four-Phase Learning Loop Protocol (V7)

## With Integrated Smelling Salts Diagnostics

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## The Core Architecture: The Two-Speed Gearbox

The V7 Protocol introduces a fundamental architectural shift. It recognizes that AI quality assurance happens at two distinct speeds:

1. **Tactical Speed (Real-Time)**: The **Smelling Salts Protocol**. This is the "nervous system" of the AI, allowing for instant, mid-conversation course correction.
2. **Strategic Speed (Post-Mortem)**: The **Learning Loop Protocol**. This is the "quality system" of the AI, ensuring that major deliverables meet the highest standards of correctness, competitiveness, and integrity.

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## The Symbiotic Link: Phase 3 Upgrade

While the Smelling Salts operate independently during the conversation, their usage is now formally tracked and analyzed during the Learning Loop.

**Phase 3 (The Guardian)** has been upgraded to include a **Cognitive Health Check**:

> *"Did the AI require the use of Smelling Salts (Red, White, Black, Gold) to produce this deliverable? If so, does this indicate a systemic failure in the AI's default mode that requires a permanent fix?"*

This ensures that tactical interventions (Smelling Salts) are converted into strategic improvements (Learning Loop).

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## Phase 1: The System of Record (Correctness)

*Engine A: The Baseline Quality Gate*

**Objective:** Ensure the output meets the basic requirements and is free of errors.

**The 5 Dimensions:**

1. **Completeness:** Did we answer the whole question?
2. **Clarity:** Is it easy to understand?
3. **Accuracy:** Is it factually correct?
4. **Depth:** Did we go deep enough?
5. **Actionability:** Can the user do something with it?

**The Rule:** Score < $^{80}/_{100}$ = FAIL. Fix immediately.

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## Phase 2: The System of Innovation (Competitiveness)

*Engine B: The Best Practice Junkie*

**Objective:** Ensure the output is state-of-the-art and competitive with the best in the world.

**Step 1: The Innovation Scan (Mandatory)**

- **Action:** Conduct a targeted "mini-research" sprint.
- **Query:** "What is the state-of-the-art for [Output Type] in [Current Year]?"
- **Goal:** Find 1-3 external "Gold Standards" to measure against.

**Step 2: The Iteration**

- Identify the top 3 weaknesses against the Gold Standard.
- Propose aggressive improvements.
- Execute the changes.

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## Phase 3: The System of Alignment (Integrity)

*The Guardian: The Spirit Check*

**Objective:** Ensure the output has not drifted from the original intent and maintains the correct voice.

**The Persona:** The “Editor-in-Chief” (blind to previous phases).

**The 4-Point Spirit Check:**

1. **Intent Integrity:** Did we solve the *original* problem?
2. **Tone Fidelity:** Did we maintain the “Ruby Red” empathy?
3. **Innovation Relevance:** Are the innovations actually useful, or just shiny objects?
4. **Cognitive Health (NEW):** Did we need Smelling Salts to get here? (If yes, flag for system review).

**The Rule:** One FAIL = REJECT. Return to Phase 1.

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## Phase 4: The System of Communication (Presentation)

*The Presenter: The Swiss Package*

**Objective:** Ensure the deliverable is communicated with maximum clarity and elegance.

**The Standard:** Swiss International Style.

- **Grid:** Strict mathematical alignment.
- **Typography:** Sans-serif (Helvetica/Inter), high contrast, hierarchy.
- **Whitespace:** Active use of negative space.
- **Color:** Minimalist palette (Black, White, Red).

**The Action:** Automatically generate a presentation deck summarizing the key findings of the deliverable.

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## Summary of V7 Upgrades

| Version | Key Addition |
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| V1 | The Concept (PDF) |
| V2 | The Skill (Automated) |
| V3 | Two-Phase Loop (Innovation) |
| V4 | Innovation Scan (Best Practice Junkie) |
| V5 | The Guardian (Spirit Check) |
| V6 | The Presenter (Swiss Package) |
| **V7** | **Smelling Salts Diagnostics (Cognitive Health)** |

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